

# Power Ledger: Delivers blockchain- powered peer- to-peer energy trading with Google Cloud

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

ABOUT POWER LEDGER

With Google Cloud, Power Ledger is delivering a blockchain-powered platform with infrastructure costs 30% below previous cloud prices. Infrastructure management has been reduced from one to five hours across development teams.



Power Ledger

## About Power Ledger

Power Ledger is a technology start-up that has developed a blockchain-enabled energy trading platform with the aim of making energy markets more efficient. The business helps people transact energy, trade environmental commodities, and transact in renewables. Power Ledger has received global recognition for its technology, and its software is being used in countries such as Australia, India, Japan, Thailand, and the United States.

## Google Cloud results

- Cuts time to manage infrastructure from one FTE to an average of five hours across development team
- Completes migration using internal resources thanks to ease of use
- Gains confidence in infrastructure support for new products and features

Reduced  
by 30%  
compute  
provisioning

Industries: Energy, Chemicals & Utilities

Location: Australia

(<https://cloud.google.com/appengine/>), [Cloud](https://cloud.google.com/) Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/functions/>), [Cloud](https://cloud.google.com/kubernetes-engine/) Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>)

[Cloud SQL for PostgreSQL](https://cloud.google.com/sql/docs/postgres/) (<https://cloud.google.com/sql/docs/postgres/>)

[App Engine](https://cloud.google.com/appengine/) (<https://cloud.google.com/appengine/>)

[Tell us your challenge. We're here to help.](https://cloud.google.com/functions/) [Cloud Functions](https://cloud.google.com/functions/) (<https://cloud.google.com/functions/>)

[Contact us / Contact](#)

[Cloud Storage](https://cloud.google.com/storage/) (<https://cloud.google.com/storage/>)

[Cloud Pub/Sub](https://cloud.google.com/pubsub/) (<https://cloud.google.com/pubsub/>)

Australia's abundance of sunlight is transforming the nation's energy system. Businesses and individuals are increasingly installing solar voltaic panels on the roofs of premises and homes to generate electricity. [According to](https://www.cleanenergycouncil.org.au/resources/technologies/solar-energy)

(<https://www.cleanenergycouncil.org.au/resources/technologies/solar-energy>)

the Clean Energy Council, a peak body for the clean

energy industry in Australia, small-scale solar—systems up to 100kW—accounted for 19.6% of Australia’s clean energy generation and produced 4.2% of the country’s total electricity.

“As the batteries used to store solar energy become more affordable, people will increasingly replace energy from the electricity grid with energy generated by their own solar installations,” explains John Bulich, Technical Director and co-founder of Power Ledger (<https://www.powerledger.io/>).

“Unfortunately, this makes maintaining grid infrastructure more expensive for people who do not have the money to invest in solar and batteries.”

With solar power uptake increasing in Australia and worldwide, tech entrepreneur Dr Jemma Green began looking at ways of enabling businesses and individuals to share energy generated by solar panels via the electricity grid. “We wanted to provide incentives to keep customers on the grid and opted for a ‘sharing economy’ model that allowed individuals and businesses to sell the surplus energy they generated from solar on the grid,” says Bulich.

[Power Ledger] Cutting 30% of their...



## A growing business

The founders then started exploring technologies and bringing on board technical specialists to help them realize their ambitions. In the three years since being founded, Power Ledger has grown to a 25-person business that specializes in building software for the energy sector. Power Ledger's key product supports peer-to-peer energy trading, and Bulich describes it as "a data processing marketplace that inputs readings from smart meters on people's houses. We send these readings to our API, and from there we process that data to identify who used what energy from what solar installation."

In some cases, individuals and businesses bid into a live marketplace to trade energy based on price, whereas in other cases Power Ledger supports trading logic that enables participants to, for example, prioritize where they purchase energy. “For example, I may want to purchase energy from a solar installation owned by my mother before buying from an installation run by a local school,” says Bulich.

Power Ledger also provides marketplace trading products for the ownership of assets, such as shares of solar farms, as well as carbon credits. “We take the same data from smart meters and generate carbon credits or renewable energy certificates based on the individual’s energy usage,” says Bulich.

The business quickly won interest from governments, individuals, and businesses and started operations with a conventional physical infrastructure. However, it quickly realized it was impractical to attempt to run versions of its software on physical infrastructure in every jurisdiction clients or prospective clients were based. In addition, managing and maintaining servers, storage, networks, and associated systems meant its lean team could not focus its energies on innovating and moving quickly.

“We found the Google Cloud user interface to be more intuitive than the alternatives and organized in a way you would expect. The product is cleaner, more transparent and includes services such as built-in logging.”

*—John Bulich, Technical  
Director and co-founder,  
Power Ledger*

## **Reviewing managed Kubernetes**

Power Ledger opted for the Kubernetes open source system to automate the deployment, management, and scaling of containerized applications and ran an unmanaged version in a multinational cloud service for about a year. However, to further relieve the

administration and management burden on its team, the business began reviewing managed Kubernetes services. “We looked at providers, and the cost and developer experience provided by Google Cloud made it the best fit for our business,” says Bulich. “Google Cloud had extensive online support and documentation for Kubernetes and its range of services.”

“We also found the Google Cloud user interface to be more intuitive than the alternatives and organized in a way you would expect,” he adds. “The product is cleaner, more transparent and includes services such as built-in logging.”

“We had easy  
access to the  
online resources  
we needed for  
the project,  
which is a good  
indication of the  
developer  
support that is  
out there for  
Google Cloud.  
Ease of use and  
lack of

complexity of  
the product also  
made our  
experience in  
undertaking the  
migration  
extremely  
positive.”

*—John Bulich, Technical  
Director and co-founder,  
Power Ledger*

Power Ledger formally decided to move to Google Cloud in December 2018 and migrated all services—including production environments—by the following February. The business completed the migration using internal resources working with online documentation and developer communities. “We had easy access to the online resources we needed for the project, which is a good indication of the developer support that is out there for Google Cloud,” says Bulich. “Ease of use and lack of complexity of the product also made our experience in undertaking the migration extremely positive.”

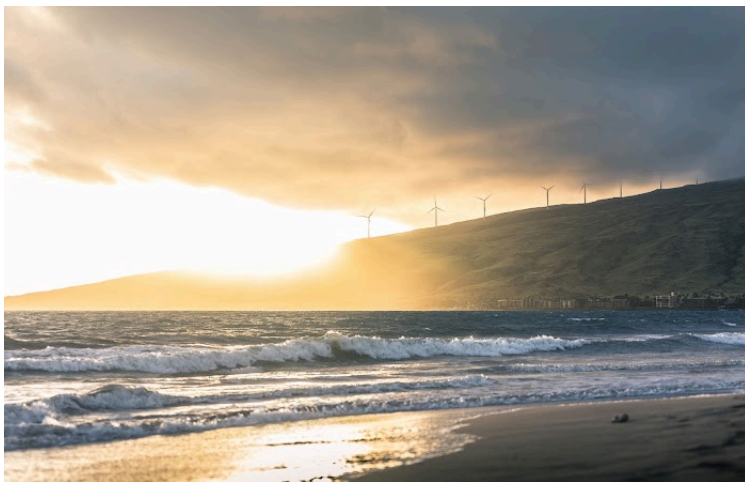
The two-month migration incorporated post-move modifications and a month running systems in parallel in both cloud services with the same data inputs to ensure they were operating in the same way. This ensured Power Ledger was recording the



same outputs from both systems before completing the move.

Power Ledger's move to Google Cloud delivered immediate success. "Since we completed the migration, we've had no issues, no downtime, and cheaper bills," says Bulich. "Not a single bill has reached the level of bills incurred at our previous cloud provider, even though we're increasing the size of our infrastructure every week."

To deliver its services, Power Ledger is using [Google Kubernetes Engine](#) (/kubernetes-engine); [Cloud SQL for PostgreSQL](#) (/sql/docs/postgres); [App Engine](#) (/appengine); [Cloud Functions](#) (/functions); [Cloud Storage](#) (/storage); [Pub/Sub](#) (/pubsub); and [Stackdriver](#) (/stackdriver).



## Positioned for growth

With Google Cloud, Power Ledger is well positioned to grow against its key metrics—the number of megawatt hours and the number of smart meters supported on its platform. "We've proven through

stress testing we can handle more than hundreds of thousands of meters per client, while all our response times are in milliseconds,” says Bulich. For users, the platform can continue to help deliver savings and information about energy usage to improve sustainability.

“Anyone who understands a cloud platform can easily pick up on Google Cloud. Because there’s effectively no barrier to entry, recruiting developers who can build on the platform is very easy. All of our developers use it without any issues.”

—John Bulich, Technical  
Director and co-founder,  
Power Ledger

According to Bulich's calculations, Power Ledger's costs are about 30% lower on Google Cloud relative to its previous cloud service. In addition, the development team has reduced the time spent on infrastructure management from the equivalent of one full-time role to an average of five hours per week. "This has freed up key resources to build more features for our products," he says.

Power Ledger's prominence as a start-up using Google Cloud has prompted queries about the ease of hiring developers and engineers skilled in the product. Bulich finds these questions easy to answer. "Anyone who understands a cloud platform can easily pick up on Google Cloud," he says. "Because there's effectively no barrier to entry, recruiting developers who can build on the platform is very easy. All of our developers use it without any issues."



## Trouble-free releases

Using Google Cloud has given the business confidence that its infrastructure and technology stack will enable the trouble-free release of new products or features. “The stability we achieve with Google Cloud makes our technology considerably more reliable,” says Bulich.

Bulich sums up Google Cloud as providing a simple development experience that allows Power Ledger to spend less time on infrastructure and more on adding value to clients. “We’re still experimenting with Google Cloud, and AI and machine learning is of interest down the track,” he says. “We have real-time data for energy customers all over the world, and that is a great starting point for our AI initiatives. We also plan to start adding features like predicting energy usage and forecasting market prices so people can take actions like charging batteries ahead of time.”

**Learn  
More**

Reach out to  
our team to  
see how  
Google Cloud  
can help your

business.